Qn	Ans	Solution
1	A	$$g = \frac{2h}{t^2}$$ $$\frac{\Delta g}{g} = \frac{\Delta h}{h} + 2\frac{\Delta t}{t}$$ $$2\% = \frac{\Delta h}{h} \times 100\% + 2 \times \frac{0.0008}{0.4000} \times 100\%$$ $$\frac{\Delta h}{h} \times 100\% = 2\% - 2\frac{0.0008}{0.4000} \times 100\%$$ $$\frac{\Delta h}{h} \times 100\% = 1.6\%$$ (common mistake: $\frac{\Delta h}{h} = \frac{\Delta g}{g} + 2\frac{\Delta t}{t} \Rightarrow \frac{\Delta h}{h} \times 100\% = 2\% + 2\frac{0.0008}{0.4000} \times 100\% = 2.4\%$)
2	B	Estimate diameter ~ 20 cm = 0.20 m Volume $\propto r^3 = 0.10^3 = 1 \times 10^{-3}$ (Volume $= \frac{4}{3}\pi(0.10)^3 = 4 \times 10^{-3}$ m^3)